

Hedging Bitcoin with Its Own Fear Gauge

Volatility Forecasts, the Volmex BVIV Index and What 15-Minute VWAP Timing Is Worth

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Abstract

Bitcoin implied volatility has become directly tradable: perpetual futures on the Volmex BVIV index have listed on Bitfinex since 2024 and on Hyperliquid since September 2026. We ask how BTC volatility should be forecast when a 30-day implied index is observable, and whether a long-BVIV position timed by a VWAP signal on 15-minute bars hedges spot or perpetual BTC exposure better than a minimum-variance hedge. In a calibrated market where the BVIV perpetual is priced without arbitrage, we evaluate seven forecasting models and 87 hedging rules with strict separation between training and 200 test paths. Implied volatility is the best forecast at 30 days, HAR-type models at one day, and HAR augmented with implied variance is robust across horizons. A minimum-variance BVIV hedge cuts expected shortfall by 4.2% for 0.88% of notional a year; volatility hedges stay partial because BTC's spot-vol correlation is weak and changes sign across regimes. Switching the hedge on and off around VWAP destroys value through turnover. A VWAP *ratchet*, which keeps a minimum-variance core and adds protection on breakdowns, lifts the ES reduction to 5.0%. A timing placebo shows what drives that gain. VWAP breakdowns do carry information: rules that hedge only after a breakdown beat randomly timed copies by up to 1.36 percentage points. But such rules protect less and cost more than a core hedge, and once the core is held, the timing value vanishes and the gain is a size effect. The placebo is the yardstick for any intraday hedge signal, and our pipeline applies it unchanged to live Binance, Bitfinex and Volmex data.

1 Introduction

A holder of bitcoin, whether in spot or through a perpetual swap, is exposed to two kinds of risk: an ordinary random walk, and sudden crashes in which liquidity disappears and implied volatility spikes. Delta hedges with futures remove the first and, with it, the position itself. Crash protection has historically required options. A cheaper and more modular instrument now exists. The Volmex BVIV index measures 30-day BTC implied volatility in a model-free way (Volmex Labs). It has traded as a USDT-margined perpetual on Bitfinex since April 2024 (CoinDesk, 2024), on gTrade since June 2025 (CoinDesk, 2025a), and as `mkt:BIV` on Hyperliquid since 21 September 2026 (CoinDesk, 2026). Regulated alternatives followed: CME announced Bitcoin Volatility futures in May 2026 (CME Group, 2026).

Whether a long-volatility position hedges a long-BTC book depends on the spot-vol correlation. In BTC this correlation is weak and unstable. Positive return shocks have historically raised volatility more than negative ones (Bouri et al., 2017; Baur and Dimpfl, 2018), an asymmetry that has faded over time (Takaishi, 2021). The correlation between BTC and DVOL turned strongly positive in early 2023 (CoinDesk, 2023), while BVIV trended down through the 2024–25 rally (CoinDesk, 2025b). Equity investors learned that long-volatility hedges look good *ex ante* but are eroded by carry and costs (Whaley, 2013; Alexander et al., 2016; Israelov, 2019), except in genuine crises (Szado, 2009). This raises the question of whether a hedge can be *timed*. This paper tests a timing signal that every intraday trader already watches, price relative to its volume-weighted average, and asks whether it carries information about crash risk beyond simply holding more hedge.

Contributions. (i) We compare seven daily volatility forecasts, including the BVIV-implied forecast, with robust loss functions. (ii) We derive the P&L of a BTC book hedged with BVIV perpetuals, splitting funding into an arbitrage-free *drift transfer* and a *carry premium*. (iii) We design a VWAP ratchet, a hysteresis rule that is fast to protect and slow to un-hedge, show why naive VWAP switching fails, and introduce a *timing placebo* that separates timing information from hedge size. (iv) We report Monte Carlo evidence under a strict train/test protocol and ship a pipeline that reproduces every number on live data.

2 The BVIV index and the economics of a volatility hedge

Index and instruments. BVIV is a model-free, 30-day constant-maturity implied-volatility index in the variance-swap tradition (Volmex Labs; Demeterfi et al., 1999). Secondary sources list Deribit and OKX

options as inputs. It updates about once per second and is served at 1- to 60-minute resolution by the Volmex API. Deribit’s DVOL uses the same VIX-style construction on Deribit options alone (Deribit, 2021), which makes it a natural substitute for history before 2024. The perpetuals pay one dollar per index point per contract. On Bitfinex the mark price is the index; funding follows the spread between perpetual and mark and can briefly pause trading. On Hyperliquid, funding adds a default interest leg of 0.01% per eight hours to a premium term (Hyperliquid, 2026). That is about 11% a year of notional, or roughly 6 vol points a year at $\text{bviv} \approx 55$. This leg is our baseline *carry* of a long-volatility position; builder-deployed markets can reconfigure it, so we vary it in Section 6.

Variance risk premium. Implied variance exceeds subsequent realised variance on average. For BTC the premium is about 0.14 a year in variance units, against about 0.02 for the S&P 500 (Almeida et al., 2024), which is consistent with Alexander and Imeraj (2021) and the equity evidence of Carr and Wu (2009). A hedge that is long bviv pays this premium indirectly. The index itself has no drift, but the perpetual’s funding prices in the index’s expected path.

Spot-vol dynamics. The minimum-variance hedge ratio (Johnson, 1960; Ederington, 1979) of BTC against bviv is proportional to their correlation. Because that correlation is regime-dependent (Huang et al., 2022), static ratios are misspecified, and dynamic estimates (Kroner and Sultan, 1993; Engle, 2002) are the baseline.

3 Forecasting BTC volatility

Measurement. From 15-minute log returns $r_{d,i}$ (96 per UTC day) we compute realised variance $RV_d = \sum_i r_{d,i}^2$, together with bipower variation and semivariances (Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2004; Patton and Sheppard, 2015). Fifteen minutes is coarser than the 5-minute rule of thumb (Liu et al., 2015), but it matches the trading frequency of the hedge and is robust to the microstructure of a 24/7 market. We use close-to-close returns and not bar VWAPs, because returns between interval averages are smoothed by about one third (Working, 1960).

Models. Each model forecasts the average daily variance over days $t+1, \dots, t+h$, $h \in \{1, 7, 30\}$, using information up to the close of day t . The models are:

- a random walk (RW);
- RiskMetrics EWMA (J.P. Morgan/Reuters, 1996);
- GARCH(1,1) with Student- t errors, re-estimated monthly;
- the HAR model of Corsi (2009) with a 7-day week;
- bviv itself, both raw ($IV_t^2/365$, “IV-raw”) and bias-corrected by OLS (“IV”);
- HAR augmented with implied variance (HAR-IV):

$$\overline{RV}_{t+1:t+h} = \beta_0 + \beta_d RV_t + \beta_w \overline{RV}_{t-6:t} + \beta_m \overline{RV}_{t-29:t} + \beta_{iv} IV_t^2/365 + \varepsilon_{t+h}. \quad (1)$$

The regressions are re-estimated daily on an expanding window that contains only targets already observed, with no overlap leakage. We evaluate forecasts with QLIKE, $L(RV, F) = RV/F - \ln(RV/F) - 1$, which is robust to proxy noise (Patton, 2011). We also report the Mincer–Zarnowitz R^2 and Diebold–Mariano tests against HAR with Newey–West variance (Mincer and Zarnowitz, 1969; Diebold and Mariano, 1995).

Results. Table 1 and Figure 1 confirm the pattern documented on Deribit data by Hoang and Baur (2020):

- At one day, the HAR family is best. HAR-IV is marginally ahead of HAR (median loss ratio 0.99) and beats it significantly on only 22% of paths. Implied volatility alone is clearly worse (1.18), because a 30-day index cannot track tomorrow’s variance, and GARCH on daily returns lags (1.26), as Bergsli et al. (2022) found for BTC.
- At seven days, the bias-corrected index and HAR-IV tie (0.90 and 0.89 of HAR’s loss).
- At 30 days, the index is best (0.81) and beats HAR significantly on 34% of paths, never losing (0%).

HAR-IV is never far from the best (0.99, 0.89, 0.90), so we use it as the 30-day forecast for hedging. Raw IV is biased upward by the variance risk premium, and the OLS bias correction removes most of that cost (0.90 versus 0.81 at 30 days).

4 A BTC book hedged with BVIV perpetuals

P&L. The book is long q BTC at price S_t and long $H_t \geq 0$ bviv perpetual contracts at mark F_t , each paying \$1 per index point. A position chosen at the close of bar t earns over bar $t+1$:

$$\Pi_{t+1} = q(S_{t+1} - S_t) - \mathbb{1}_{\text{perp}} q S_t f_t \Delta + H_t (F_{t+1} - F_t) - H_t \phi_t - c |H_{t+1} - H_t| F_{t+1}, \quad (2)$$

Model	$h = 1$ days			$h = 7$ days			$h = 30$ days		
	QL ratio	MZ R^2	DM win	QL ratio	MZ R^2	DM win	QL ratio	MZ R^2	DM win
RW	1.07	0.16	7%	1.33	0.17	0%	1.39	0.08	1%
EWMA	1.38	0.08	0%	1.31	0.10	0%	1.48	0.08	0%
GARCH	1.26	0.10	0%	1.13	0.10	0%	1.00	0.04	4%
HAR	1.00	0.17	–	1.00	0.15	–	1.00	0.04	–
IV-raw	1.22	0.18	0%	0.99	0.29	11%	0.90	0.22	10%
IV	1.18	0.18	2%	0.90	0.27	24%	0.81	0.16	34%
HAR-IV	0.99	0.22	22%	0.89	0.27	32%	0.90	0.13	22%

Table 1: Out-of-sample forecast accuracy relative to HAR across 200 test paths. QL ratio: median ratio of a model’s QLIKE to HAR’s (below 1 is better; best in bold). MZ R^2 : mean Mincer–Zarnowitz R^2 . DM win: share of paths on which a Diebold–Mariano test rejects equal accuracy in the model’s favour at the 5% level.

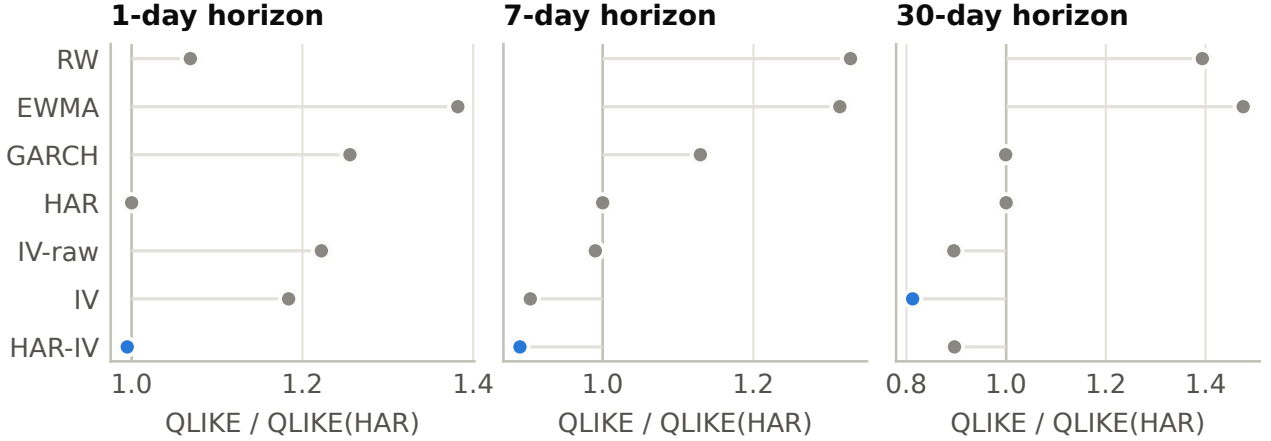


Figure 1: Median QLIKE relative to HAR by model and horizon; the best model at each horizon is highlighted.

where f_t is the BTC perpetual’s funding rate, ϕ_t the BVIV funding per contract and c the proportional trading cost. A perpetual with funding intensity k trades at $\mathbb{E}_t[\int_0^\infty k e^{-ks} I_{t+s} ds]$, the expected index sampled at a random time (He et al., 2022; Ackerer et al., 2025). In the fast-funding limit the mark equals the index, and funding splits into two parts:

$$\phi_t = \underbrace{\mathbb{E}_t[I_{t+1}] - I_t}_{\text{drift transfer}} + \underbrace{\bar{\kappa} \Delta}_{\text{carry premium}}. \quad (3)$$

The first term compensates longs exactly for the index’s expected path: after a spike, longs *receive* funding while the index mean-reverts. Only the premium $\bar{\kappa}$ (vol points a year) is a cost, so we report hedge costs as carry premium plus trading costs, not as raw funding.

Hedge ratios. The minimum-variance hedge is $H_t^* = -qS_t \beta_t$, where $\beta_t = \text{Cov}_t(r, \Delta F) / \text{Var}_t(\Delta F)$ is estimated on 15-minute changes with an EWMA (half-life 14 days). We impose $H \geq 0$, since a protective hedge is never short volatility. For crash sizing we also estimate a *downside hedge ratio* $\beta_t^- = \mathbb{E}_t[r \Delta F \mid r < 0] / \mathbb{E}_t[\Delta F^2 \mid r < 0]$ on falling bars only, shrunk towards β_t by its effective sample size, which gives $H_t^- = -qS_t \beta_t^-$. The moments are deliberately not demeaned, so β^- minimises the downside second moment of the hedged book and also offsets the average loss on falling bars.

The VWAP signal. BTC trades 24/7, so we anchor VWAP on a rolling 24-hour window, which avoids an arbitrary session open; Section 6 also tests a UTC-session anchor. Exchange klines give each bar’s exact VWAP as quote volume divided by base volume. For Brownian prices, the gap between the last price and its average over a window w has standard deviation $\sigma\sqrt{w/3}$, so the scale-free breakdown statistic is

$$z_t = \frac{\ln(S_t / \text{VWAP}_t^{24h})}{\hat{\sigma}_t \sqrt{1/3}}, \quad (4)$$

where $\hat{\sigma}_t$ is a causal one-day volatility forecast. It is an EWMA of 15-minute squared returns, de-seasonalised by a trailing hour-of-week profile; intraday volatility in crypto follows strong and growing periodic patterns (Andersen and Bollerslev, 1997; Hansen et al., 2024; Eross et al., 2019). In this way the volatility forecast of

Moment	Median	5–95%	Empirical anchor
Realised vol, annualised (%)	51.5	[45.1, 59.4]	calibration target ≈ 50
Mean BVIV (vol pts)	57.1	[52.3, 63.1]	35.5 (Aug 2026) to >96 (Feb 2026)
IV – subsequent 30d RV (vol pts)	7.6	[4.0, 10.6]	VRP ≈ 0.14 p.a. in variance
Share of days IV $>$ RV (%)	78	[66, 87]	VRP positive on average
corr(daily return, Δ IV)	−0.21	[−0.28, −0.12]	sign varies by regime
corr on falling 15m bars	−0.26	[−0.34, −0.20]	IV spikes in sell-offs
corr on rising 15m bars	0.03	[−0.02, 0.09]	IV up in 2023, down in 2025 rallies
Excess kurtosis, daily returns	3.7	[1.7, 10.0]	fat tails
Worst daily loss (%)	15.0	[10.0, 25.3]	fat left tail

Table 2: Simulated moments across 200 test paths of 900 days (15-minute bars).

Section 3 sets the band width. Intraday momentum and reversal both exist in BTC (Shen et al., 2022; Wen et al., 2022), so we treat z as a *state* variable, not a return forecast.

Rules. We compare four families, each with long-only positions, a hedge cap of 50% of BTC notional and a rebalancing band $b = 25\%$. The band follows the no-trade-region logic of Whalley and Wilmott (1997): the hedge is only traded when its target moves more than b away from the current position.

- **Always-on:** $H_t = \lambda H_t^*$, scaled by $\lambda \in \{0.5, \dots, 3\}$.
- **VWAP-switch:** $H_t = g_t H_t^-$. The gate g_t turns on when $z_t < z_{in}$ and off when $z_t > z_{out}$, after a minimum hold of four hours.
- **VWAP-ratchet:** an overlay o_t jumps to 1 on a breakdown and decays with half-life τ :

$$o_t = \begin{cases} 1 & z_t < z_{in} \\ 2^{-1/\tau} o_{t-1} & \text{otherwise,} \end{cases} \quad H_t = \varphi H_t^* + o_t \max(H_t^- m_t - \varphi H_t^*, 0), \quad (5)$$

where φ is a core floor and m_t an optional size multiplier that grows with forecast “cheapness”, the log ratio of the HAR-IV variance forecast to implied variance.

- **Oracle:** H_t^- held only during the latent stress regime. It is infeasible and serves as an upper bound for regime timing.

The ratchet’s asymmetry is the point of the design. Re-triggering inside an episode refreshes o_t without trading, so whipsaw around VWAP costs nothing, whereas the switch pays a round trip every time price crosses the band.

5 Experimental design

Calibrated market. Live exchange data could not be accessed from the build environment, so we evaluate the rules in a simulated market calibrated to the stylised facts above. The pipeline nevertheless ships loaders for live data (Section 7).

- *Variance and returns.* Log-variance is the sum of a slow and a fast Ornstein–Uhlenbeck factor. The fast factor tracks a three-state hidden regime (calm, stress, euphoria; expected durations 30/5/10 days) and receives EGARCH-type shocks (Nelson, 1991), $u_t = \theta_R z_t + \gamma(|z_t| - \sqrt{2/\pi}) + \sqrt{1 - \theta_R^2 - \gamma^2(1 - 2/\pi)} \varepsilon_t$. Here z_t is the return shock. The signed loading θ_R is negative in calm and stress and positive in euphoria (inverse leverage), and the size loading γ makes large moves of either sign raise volatility.
- *Jumps and seasonality.* Compound-Poisson jumps cluster in stress and push variance up. Volatility and volume follow hour-of-day and weekend seasonality, and bars are built from one-minute paths so that OHLC and the bar VWAP behave like exchange klines.
- *Index.* $bviv$ equals the model’s own expectation of 30-day average variance, inflated by a variance premium and perturbed by a persistent sentiment factor and fixing noise. The expectation is exact over regime paths: a Feynman–Kac product of matrix exponentials of $Q + \text{diag}(V)$ integrates the fast factor’s regime-driven drift and jump kicks, and the jump variance of returns is mixed with e^{Qs} . A unit test checks it against brute-force simulation.
- *Funding.* The drift transfer in (3) is computed exactly one bar ahead: summed over regime transitions, by Gauss–Hermite quadrature over jumps, and to second order in Gaussian shocks. A unit test verifies it against brute-force simulation, so the perpetual offers no free lunch from mean reversion.

Table 2 compares simulated moments with their empirical anchors, and Figure 2 shows a representative path.

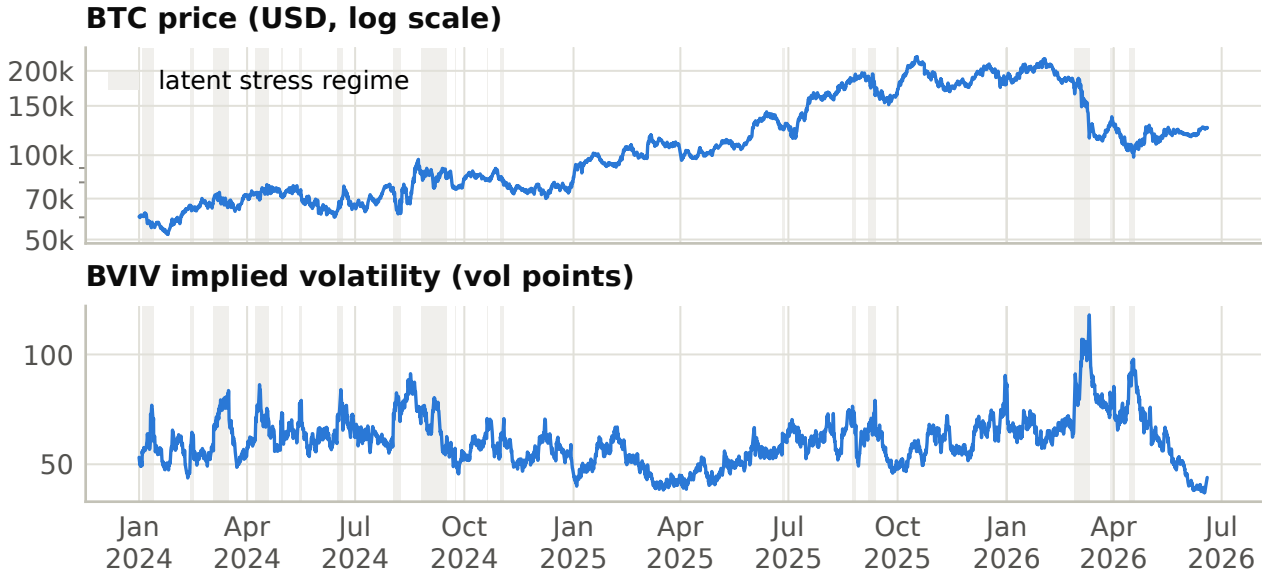


Figure 2: A representative simulated path: BTC and bviv on 15-minute bars (4-hour sampling for display), with the latent stress regime shaded.

Rule	ES red.	Var red.	Max DD	Tail offset	Carry	Trading	Turnover	Time on
Unhedged	0.0	0.0	49.5	0.0	0.00	0.00	0.0	0
Always-on (MV)	4.2	4.1	48.7	5.9	0.74	0.14	0.9	93
Always-on, 1.5×	4.9	3.1	49.5	8.8	1.11	0.20	1.3	93
VWAP-switch	2.8	1.9	50.9	4.9	0.38	3.94	24.6	35
VWAP-ratchet	5.0	3.8	48.9	7.9	1.00	0.22	1.4	98
Ratchet, placebo timing	5.0	3.8	49.0	7.7	1.00	0.22	1.4	98
Oracle (infeasible)	2.3	2.3	48.1	3.5	0.11	0.27	1.7	9

Table 3: Hedging a long-BTC spot book on 200 test paths (medians, in %). ES/Var red.: reduction in daily $ES_{97.5}$ and variance relative to unhedged. Max DD: maximum drawdown. Tail offset: hedge P&L net of costs as a share of BTC losses on the worst 5% of days. Carry and Trading: costs in % of BTC notional a year. Turnover: hedge notional traded per year as a multiple of BTC notional. Time on: share of bars with a hedge. Placebo timing: the selected ratchet with its trigger series circularly shifted (mean of 5 shifts).

Protocol. All signals are causal; unit tests verify that truncating the sample leaves every past value unchanged. Rule parameters are selected on 48 *training* paths, separate from the 200 *test* paths reported below. This avoids the data-snooping trap of picking the best of many rules on the evaluation sample (White, 2000; Hansen, 2005; Bailey et al., 2014). The grid contains 87 rules: $z_{in} \in \{-1, -1.5, -2\}$, $\tau \in \{0.5, 1, 2, 4\}$ days, $\varphi \in \{0, 0.5, 1\}$, with and without forecast sizing, plus the switch, always-on and oracle families. The selection criterion is the median reduction in expected shortfall ($ES_{97.5}$) of daily returns, measured *net* of carry and trading costs. Hedging is evaluated from day 270, once every forecast and signal exists, and forecasts from day 365. Costs are 6 bp fee plus 10 bp slippage per side on traded notional, and the carry is 6 vol points a year.

6 Results

Selected rules. On the training paths, the best ratchet keeps the full minimum-variance hedge as its core ($\varphi = 1$), triggers at $z < -1$ and decays with a half-life of 4 days. It runs with forecast-based sizing: in training the two variants were within noise of each other, and on the test paths the sized twin differs by +0.04 pp (95% CI $[-0.09, +0.17]$). The best switch enters at $z < -1$ and exits at $z > 0.5$.

Hedging performance. Table 3 summarises the test results, and Figure 3 shows the rules on one episode.

- *Always-on.* The minimum-variance hedge reduces ES by 4.2% and variance by 4.1% with a notional of only 7.0% of the BTC position. A volatility index is a partial hedge for a long-BTC book: the simulated correlation between daily returns and bviv changes is -0.21 (-0.26 on falling and 0.03 on rising bars), and the variance reduction of a minimum-variance hedge is bounded by the squared correlation (Ederington, 1979).
- *VWAP-switch.* The switch is the instructive failure. It trades 24.6 times the BTC notional per year, and its

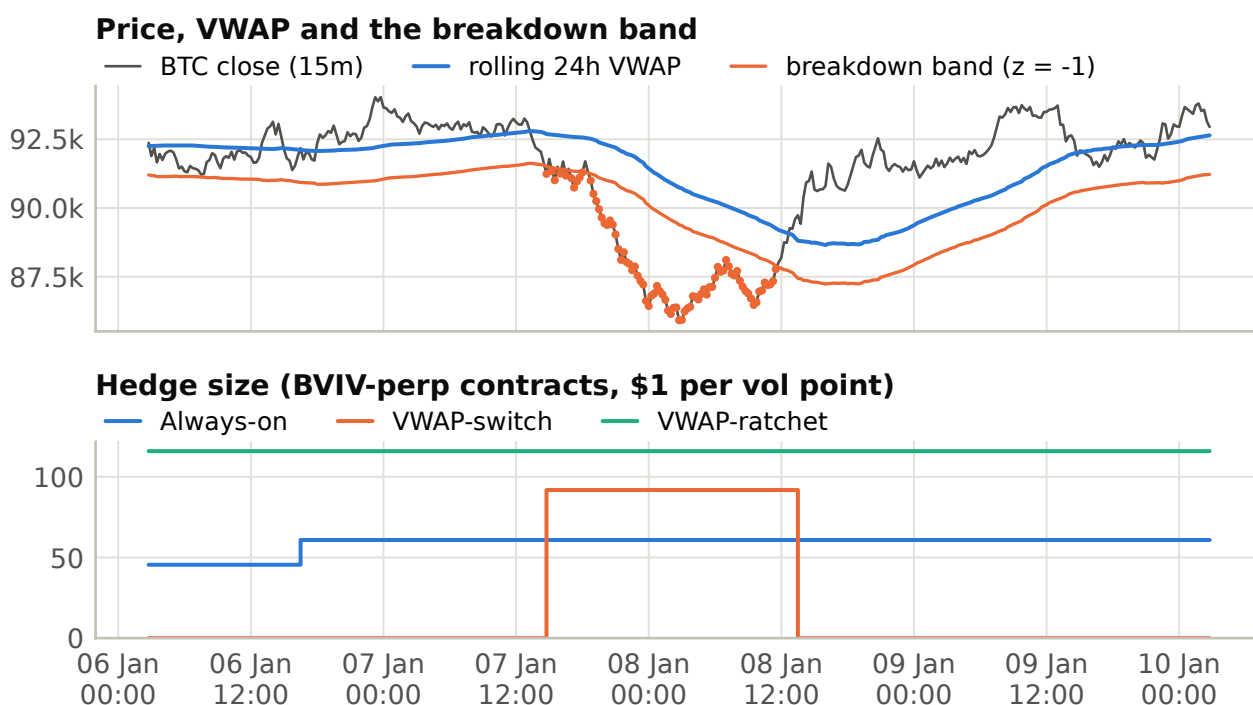


Figure 3: A fresh VWAP breakdown on the representative test path (the first trigger after three quiet days). Top: 15-minute close, rolling 24-hour VWAP and the breakdown band (orange dots: trigger bars). Bottom: hedge sizes. The switch buys protection only once the breakdown is under way and sells it as price recovers, paying a round trip. The selected ratchet barely moves: with a four-day half-life and the 25% rebalancing band, earlier triggers keep it near its upper level, so in effect it is a larger always-on hedge, which is the size effect identified by the placebo test.

ES reduction falls short of the always-on hedge by -1.67 pp (95% CI $[-2.00, -1.36]$). The gate fires on noise as often as on crashes, and each false alarm costs a round trip.

- **VWAP-ratchet.** The ratchet reduces ES by 5.0% at 1.21% a year (always-on: 0.88%). Paired across test paths, it beats the always-on hedge by $+0.61$ pp (95% CI $[+0.41, +0.81]$; higher on 67% of paths) and offsets $+2.1$ pp more of the tail losses, at the price of -0.48 pp of variance reduction.
- **Oracle.** Holding the hedge only in the latent stress regime reduces ES by just 2.3%. Most simulated crash days start from calm, so regime timing without a core hedge leaves the book exposed.

Timing or size? A placebo test. The ratchet holds more hedge on average than the always-on rule (9.6% versus 7.0% of notional), so its gain could come from size alone. To separate the two, we circularly shift each path's trigger series by 35 to 259 days, in multiples of a week. This keeps the number, clustering and hour-of-week profile of the triggers but breaks their link to the market. For the selected ratchet the placebo is just as good: it reduces ES by 5.0%, and the ratchet beats it by only $+0.02$ pp (95% CI $[-0.04, +0.07]$). Scaling the static hedge by 1.5 achieves the same ES reduction (4.9% at 1.32%), although with less variance reduction (3.1% versus 3.8%).

The grid tells a more nuanced story. Across all 72 ratchet configurations (64 paths, 3 shifts each), the timing value averages $+0.30$ pp, from -0.15 to $+1.36$ pp. 29 configurations beat their placebo at the 5% level and 0 lose to it. VWAP breakdowns therefore do carry information about tail days, but its value depends on the core hedge:

- without a core ($\varphi = 0$), timing adds $+0.69$ pp on average;
- with half a core ($\varphi = 0.5$), it adds $+0.20$ pp;
- with the full core ($\varphi = 1$), it adds 0.00 pp.

Rules that rely on timing protect less (median ES reduction 3.5% without a core, against 4.9% with one), and the correlation between a configuration's timing value and its ES reduction is -0.43 . The most timing-dependent rule reaches 4.1% at a cost of 4.2% a year, against 4.6% for the always-on hedge on the same paths. Being hedged when the breakdown comes matters more than reacting to it.

Sizing. Figure 4 places every rule in cost-protection space. Along the scaled static hedge, ES reduction peaks between $1\times$ and $1.5\times$ and then falls (4.5% at $2\times$, 1.8% at $3\times$). The extra contracts add volatility-of-volatility risk faster than they add crash convexity. The ratchet configurations scatter around this frontier,

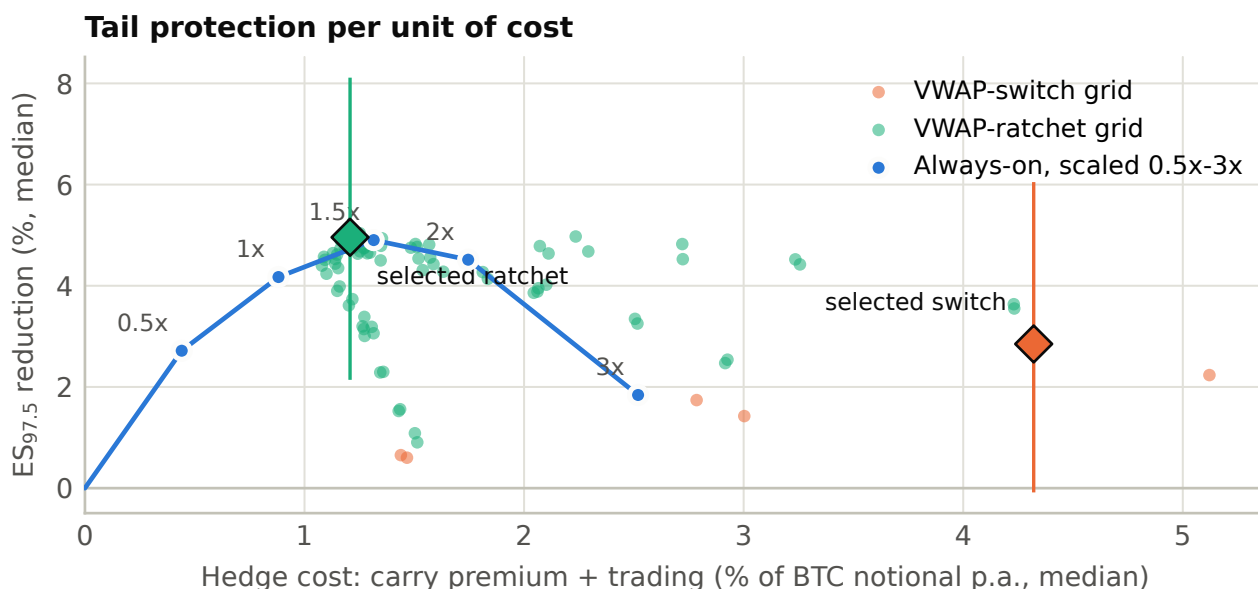


Figure 4: Tail protection versus cost on the test paths (medians). Blue: the always-on hedge scaled from 0.5× to 3×. Dots: every candidate VWAP rule; diamonds: rules selected on the training paths, with interquartile ranges of the ES reduction.

Scenario	ES red. (%)		Ratchet – Always mean [95% CI]	> 0	Switch ES red.	Cost (% p.a.)	
	Always	Ratchet				Always	Ratchet
Baseline	4.3	5.2	+0.61 [+0.29, +0.91]	74%	3.6	0.87	1.24
Trading costs x2	4.3	5.1	+0.59 [+0.27, +0.89]	73%	3.4	1.01	1.46
Trading costs x0.5	4.3	5.2	+0.62 [+0.30, +0.92]	75%	3.7	0.80	1.12
Funding carry 0	4.3	5.2	+0.62 [+0.30, +0.92]	74%	3.7	0.13	0.22
Funding carry 12	4.3	5.1	+0.60 [+0.28, +0.90]	74%	3.6	1.61	2.25
VIX-like spot-vol	9.7	10.3	+0.66 [+0.37, +0.93]	71%	6.3	1.33	1.68
Inverse leverage	0.5	0.7	+0.15 [−0.08, +0.40]	58%	0.5	0.25	0.57
Crashes cluster in stress	5.3	5.1	+0.29 [−0.08, +0.66]	57%	3.1	0.85	1.18
Session-anchored VWAP	4.3	5.2	+0.62 [+0.30, +0.92]	73%	3.4	0.87	1.25
Perpetual BTC book	4.3	5.1	+0.59 [+0.28, +0.89]	74%	3.6	0.87	1.24

Table 4: Robustness on 96 fresh paths per scenario (a separate seed block, common across scenarios), with the selected rules held fixed. ES red.: median reduction in daily $ES_{97.5}$. Ratchet – Always: mean paired difference in percentage points, with a 95% bootstrap CI and the share of paths with a positive difference. Cost: carry premium plus trading costs.

and 47% of them lie above it. The best of them do so through a better-shaped size profile, not timing: they buy the ES reduction of a 1.5× static hedge with more variance reduction and at lower cost. The economically first-order choice is how much bv_{IV} to hold.

Robustness. Table 4 varies one assumption at a time.

- *Costs and carry.* Doubling or halving trading costs, or setting the carry to zero or twelve vol points, leaves the ratchet’s edge over the always-on hedge at about +0.6 pp and significant.
- *Spot-vol regime.* The edge holds under VIX-like dynamics (+0.66 pp) but shrinks to +0.15 pp under inverse leverage (CI [−0.08, +0.40]). Here rallies lift bv_{IV} , and a hedge sized on falling bars covers less of what moves the index.
- *Crash clustering.* When crashes cluster in the stress regime, the edge halves and loses significance (+0.29 pp, CI [−0.08, +0.66]). The larger post-breakdown position then also exposes the book to volatility crashes. On the hedged book’s own worst days, the ratchet’s hedge leg returns −0.09% of notional, against +0.16% for the always-on hedge (baseline: +0.17% and +0.28%). This echoes the equity lesson that badly timed protection can deepen losses (Israelov, 2019).
- *VWAP anchor and book.* A UTC-session anchor gives +0.62 pp. Hedging a perpetual instead of spot gives +0.59 pp: BTC funding shifts the book’s mean return but not its tail co-movement with bv_{IV} .

7 Implementation on live data

The pipeline (`scripts/run_paper.py -live`) maps live data into the simulator’s schema, so forecasting, signals, hedging and reporting share one code path. It uses the following sources:

- BTCUSD 15-minute klines from the Binance bulk archive, whose quote volume over volume gives each bar’s exact VWAP, and BTC perpetual funding;
- index, mark and funding of the Bitfinex bvvv perpetual since April 2024;
- bvvv bars from the Volmex API, or DVOL as a proxy before 2024.

Four practical issues deserve attention. First, the bvvv perpetuals are young and thin, so the costs in Table 4 should be stressed further for large books. Second, oracle risk is real on builder-deployed markets: in July 2026 a single erroneous pre-market print triggered large liquidations on another Hyperliquid HIP-3 perpetual. Third, venue mechanics differ: Bitfinex warns that its funding process can briefly pause trading, so a hedge should not depend on a single venue at the moment of a crash. Fourth, the 15-minute boundary itself attracts algorithmic order flow (Kim and Hansen, 2026), so hedge orders should be worked inside the bar, for example by VWAP slicing (Berkowitz et al., 1988; Konishi, 2002; Białkowski et al., 2008), and not sent at the bar close (Madhavan, 2002).

Limitations. The evidence is simulation-based. The simulator links VWAP breakdowns to crashes only through regime drift and volatility clustering. Real liquidation cascades may make intraday breakdowns more informative, for instance through intraday momentum (Shen et al., 2022), or less. Our finding that a core hedge makes timing redundant is therefore a statement about this calibrated market, not a theorem. On live data, the placebo test of Section 6 is the decisive check, and the pipeline runs it unchanged. The forecasting comparison is cleaner on real data too, because in the simulation the index is by construction a near-rational forecast; its ranking nevertheless matches the empirical literature (Hoang and Baur, 2020; Bergsli et al., 2022).

8 Conclusion

bvvv perpetuals turn BTC implied volatility into a hedging instrument, but a partial and costly one. Four practical lessons follow.

1. Forecast BTC volatility with HAR-type models at short horizons and with the implied index beyond a week; HAR augmented with implied variance is the robust default.
2. Size a bvvv hedge with the minimum-variance ratio, or up to about 1.5 times it if the aim is expected shortfall rather than variance, and keep it on.
3. Do not switch the hedge around VWAP, because turnover eats the protection.
4. Treat a 15-minute VWAP breakdown as information that is real but secondary. It helps a hedge that would otherwise be absent, but once a core hedge is held, a ratchet’s gain is size, not timing.

Any intraday timing rule should beat a placebo with the same trigger statistics before it earns its complexity.

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